Black-Scholes Asymptotes

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Under the risk-neutral measure, with constant $r \geq 0$ and $\sigma > 0$, no dividends, $\tau = T - t > 0$, strike $K > 0$ and spot $S_t > 0$, Black-Scholes supplies:

$$C = S_t \Phi(d_1) - K e^{-r\tau} \Phi(d_2), \qquad \begin{aligned} d_1 &= A + \tfrac{1}{2}\sigma\sqrt{\tau}, \\ d_2 &= A - \tfrac{1}{2}\sigma\sqrt{\tau}, \end{aligned} \qquad A := \frac{\ln(S_t/K) + r\tau}{\sigma\sqrt{\tau}}. \tag{1}$$

Since $0 < \Phi(x) < 1$ and the subtracted term is positive,

$$0 < C < S_t \qquad \text{for all admissible parameters.} \tag{2}$$

Below, every parameter other than the one named is held fixed.

1. Volatility, $\sigma \to \inf$**.** The numerator of A is free of σ, so

$$\begin{aligned} A = O(\sigma^{-1}) \to 0, \quad \tfrac{1}{2}\sigma\sqrt{\tau} \to \infty \quad &\Longrightarrow \quad d_1 \to +\infty, \qquad d_2 \to -\infty \\ &\Longrightarrow \quad \Phi(d_1) \to 1, \qquad \Phi(d_2) \to 0 \\ &\Longrightarrow \quad C \to S_t \cdot 1 - K e^{-r\tau} \cdot 0 = S_t \qquad \blacksquare \end{aligned}$$

2. Moneyness, $S_t/K \to \inf$**.** Here $\ln(S_t/K) \to +\infty$ with $\frac{1}{2}\sigma\sqrt{\tau}$ fixed, so both arguments diverge to the same side and $\Phi(d_2) \to 1$

$$\begin{aligned} A \to +\infty \quad &\Longrightarrow \quad d_1 \to +\infty, \qquad d_2 \to +\infty \\ &\Longrightarrow \quad \Phi(d_1) \to 1, \qquad 0 < K e^{-r\tau} \Phi(d_2) \leq K \to 0 \\ &\Longrightarrow \quad C \to S_t \cdot 1 - 0 = S_t \qquad \blacksquare \end{aligned}$$

3. Maturity, $\tau \to \inf$**.** Both terms of (1) now move. Rescaling,

$$d_1 = \frac{\ln(S_t/K)}{\sigma\sqrt{\tau}} + \sqrt{\tau}\left(\frac{r}{\sigma} + \frac{\sigma}{2}\right), \qquad d_2 = \frac{\ln(S_t/K)}{\sigma\sqrt{\tau}} + \sqrt{\tau}\left(\frac{r}{\sigma} - \frac{\sigma}{2}\right), \tag{3}$$

with leading fractions $O(\tau^{-1/2})$. The bracket in d_1 is positive, that in d_2 has the sign of $r - \frac{1}{2}\sigma^2$, which is unrestricted. Hence

$$\begin{aligned} r \geq 0,\ \sigma > 0 \quad &\Longrightarrow \quad d_1 \to +\infty \quad \Longrightarrow \quad \Phi(d_1) \to 1, \\ r < \tfrac{1}{2}\sigma^2 \quad &\Longrightarrow \quad d_2 \to -\infty \quad \Longrightarrow \quad \Phi(d_2) \to 0, \\ r \geq \tfrac{1}{2}\sigma^2 \text{ (so } r > 0) \quad &\Longrightarrow \quad e^{-r\tau} \to 0, \end{aligned}$$

and the two cases combine as

$$0 < K e^{-r\tau} \Phi(d_2) \leq K \min\{e^{-r\tau},\, \Phi(d_2)\} \to 0 \quad \Longrightarrow \quad C \to S_t \cdot 1 - 0 = S_t \qquad \blacksquare$$

Collecting (2) with the three limits,

$$\lim_{\sigma \to \infty} C = \lim_{S_t/K \to \infty} C = \lim_{\tau \to \infty} C = S_t = \sup C, \tag{4}$$

the supremum being approached but never attained. With a dividend yield $q > 0$ the bound reads $C \leq S_t e^{-q\tau}$.